

Yuhang Fan

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Education

Tongji University

Master's Candidate in Physics

Sep 2024 - Jun 2027

Shanghai, China

Research Focus: Optical Metrology, Spectroscopic Ellipsometry, Thin-Film Measurement

Soochow University

Bachelor of Science in Physics

Sep 2020 - Jun 2024

Suzhou, China

Academic Performance: GPA: 3.7/4.0, Rank: 12/87 (Top 15%)

Research & Project Experience

Ellipsometry and TEM Film-Thickness Metrology

Primary Contributor

Mar 2025 - Present

- Designed and executed a comparative study of thin film thickness measurement using ellipsometry and TEM, developing independent measurement models and traceability frameworks to support reliable comparison across techniques and rigorous interpretation of measurement differences.
- Characterized measurement performance through repeatability, spatial uniformity, image analysis, and uncertainty evaluation; compared results across techniques and determined how sample uniformity, measurement procedures, and data processing affected result comparability.
- Validated agreement across measurement methods through metrological compatibility analysis and combined validated results into a robust reference value; the manuscript has been submitted to a journal indexed in SCI.

Automated Nanoscale Measurement from TEM Images

Independent Researcher

Sep 2025 - Present

- Developed a MATLAB image-processing workflow integrating denoising, contrast enhancement, FFT, Canny edge detection, and feature localization to automatically extract film thickness, lattice features, and grating period from TEM images.
- Compared centroid, half-intensity, and gradient-based measurement methods to evaluate result consistency, establishing a repeatable process for nanoscale dimensional measurement.

CNAS Accreditation and Equipment Qualification

Team Member

Sep 2025 - Present

- Aligned laboratory equipment records with CNAS accreditation criteria and national standards, standardizing instrument specifications, operating procedures, maintenance documentation, and measurement-capability evidence for accreditation review and equipment qualification.

Skills

Software: MATLAB; Python; Origin; Excel; PowerPoint; Gatan DigitalMicrograph.

Instrumentation: Spectroscopic Ellipsometry; Transmission Electron Microscope (TEM); Atomic Force Microscope (AFM).

Languages: Chinese (Native); English (CET-6).

Honors & Awards

- Jiang Zhen Scholarship, Tongji University (received twice).
- Academic Excellence Scholarship, Soochow University (received three times, best rank: 3rd in major)
- Comprehensive Scholarship, Soochow University (received twice).
- Merit Student Honor, Soochow University (received twice).
- Outstanding Graduate Honor, Soochow University.

Interests

Blindfolded Rubik's Cube Solving; Logic Puzzles; Music Theory; Economics and Financial Markets.